

Lab: Multi-Agent Inference and Communication

Learning Goal: Model communication as coupled inference and analyze synchrony.

Part 1: Coupled Generative Models

Exercise: Two agents observe the same weather but from different locations.

- Agent A is indoors (observes: room temperature changes)
- Agent B is outdoors (observes: rain, sunshine)
- Hidden state s : actual weather {sunny, rainy, cloudy}
- Write Agent A's likelihood $P_A(o | s)$ — how weather affects indoor temperature
- Write Agent B's likelihood $P_B(o | s)$ — how weather is directly observed
- If they share their observations, how does this improve each agent's inference?

Write your response here

Part 2: Measuring Generalized Synchrony

Learning Goal: Understand when agents "converge" on shared beliefs.

Scenario: Agent A and Agent B discuss whether a movie is good.

- Before conversation: A thinks 70% good, B thinks 30% good
- After sharing reasons: A thinks 60% good, B thinks 55% good
- Did generalized synchrony increase? How can you tell?

- What would perfect synchrony look like? (both posteriors identical)
- What would zero synchrony look like? (beliefs unchanged by interaction)
- Is synchrony always good? Give an example where maintaining different beliefs is valuable.

Write your response here

Part 3: Language as Model Compression

Learning Goal: Analyze how sentences compress generative models.

Exercise: For each sentence, identify what generative model information is being transmitted:

Sentence	Hidden states	Transition (B)	Observation (A)	Preferences (C)
"Watch out for the car!"				
"If it rains tomorrow, bring an umbrella"				
"The restaurant on Main St. has great pasta"				

Write your response here

Part 4: Shared Priors and Social Norms

Learning Goal: Model cultural norms as shared prior distributions.

Exercise: For each cultural norm, express it as a shared prior and explain how it reduces collective prediction error:

1. "Drive on the right side of the road" → Shared $D = ?$
2. "Be quiet in libraries" → Shared $C = ?$
3. "Shake hands when meeting" → Shared $B = ?$

What happens when someone violates the norm? (In terms of prediction error)

Write your response here

Part 5: Reflection

In 150 words, reflect: If communication is coupled inference and cultural norms are shared priors, does this mean all disagreement is "failing to minimize joint free energy"? Or is disagreement sometimes adaptive?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Multi-agent modeling	Coupled generative models
2	Synchrony analysis	Convergence of beliefs
3	Linguistic analysis	Language as model compression
4	Norm formalization	Cultural priors
5	Critical reflection	Disagreement and free energy